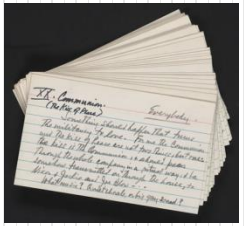
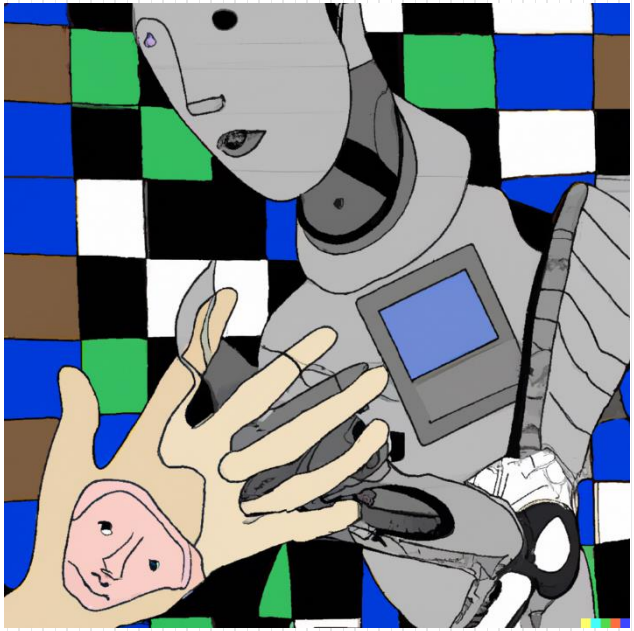


Conducting your research



Six types of people (AI)

Sceptic:

"It is just a hype"

Secret User:

"I don't tell anyone I use ChatGPT…"

Power Prompter: *"I've got a prompt for everything!"*

Copy-Paster:

"I just ask it to write a worksheet. Done."

Ethical Warrior: *"We can't use this until we talk about safety and bias."*

Curious but Cautious:

"I can see the potential…but where do I start?"

Generative Artificial Intelligence (AI)

Prompt

Gemini ChatGPT Grok Copilot

Meta AI deepseek Claude

Essay or poem

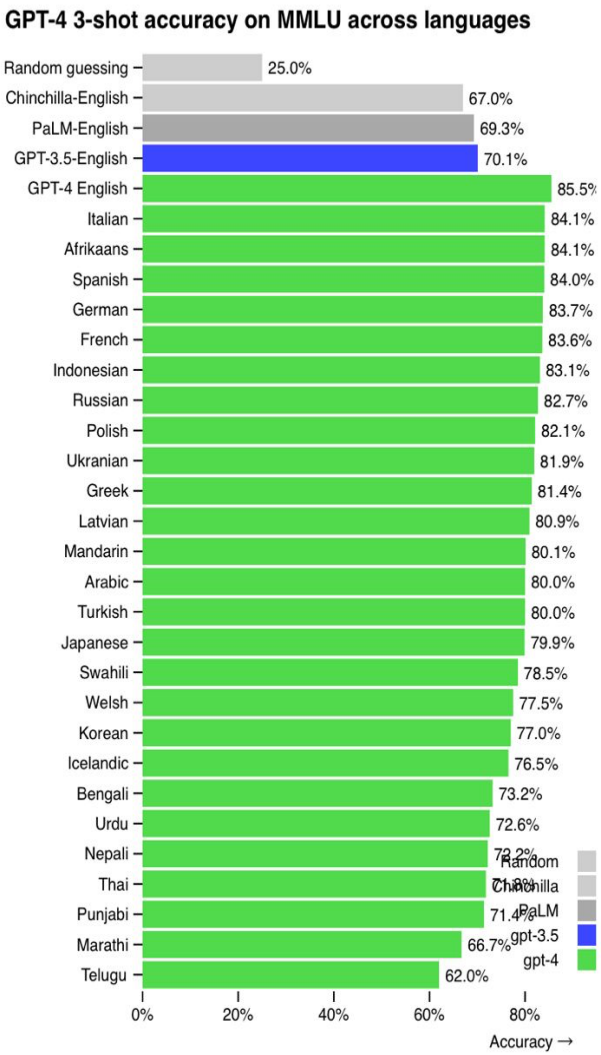
Coding

Data Analyses

Artwork

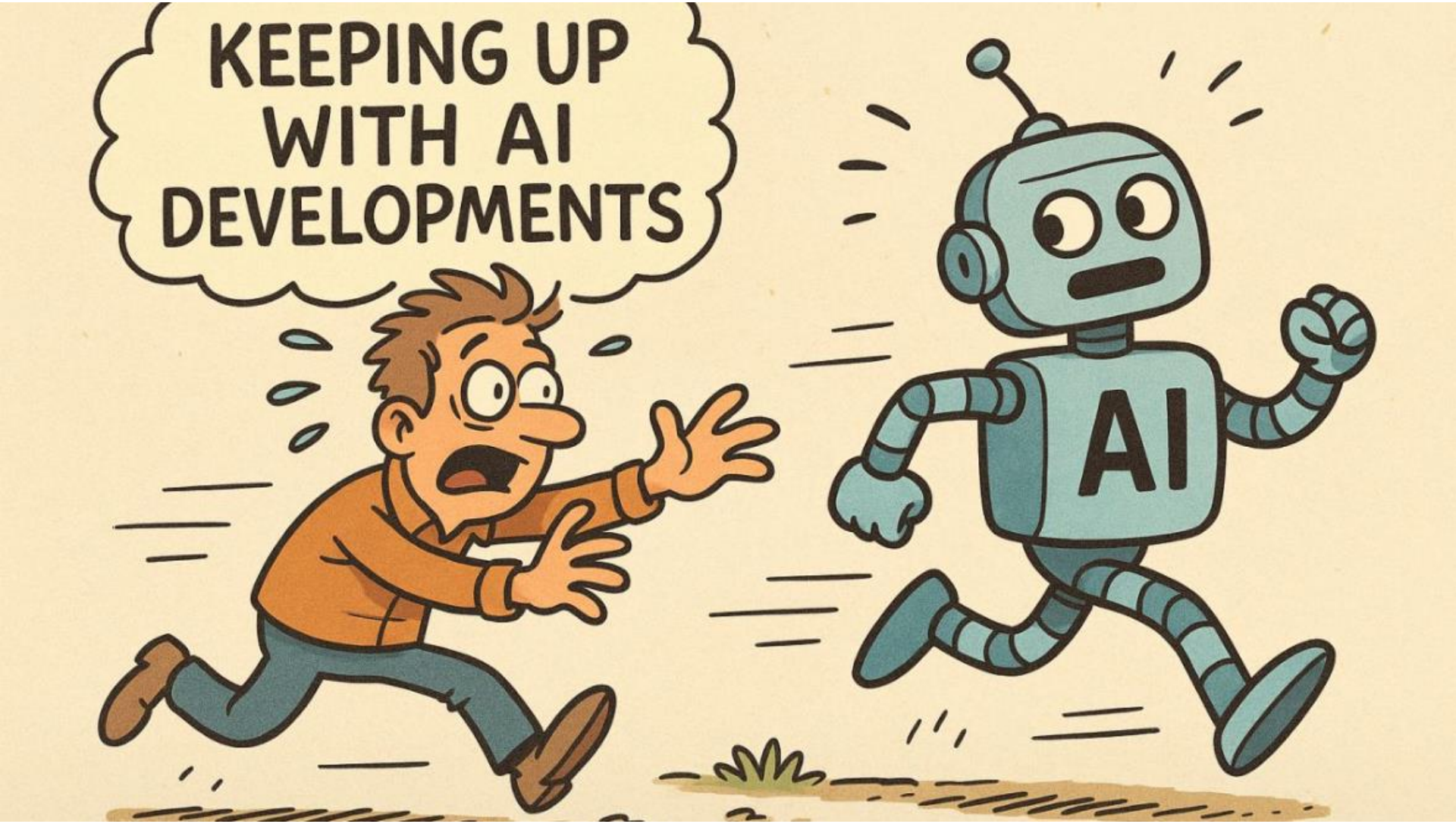
Music

ChatGPT claims its new voice mode can speak three official South African languages other than English: Afrikaans, isiZulu and isiXhosa.



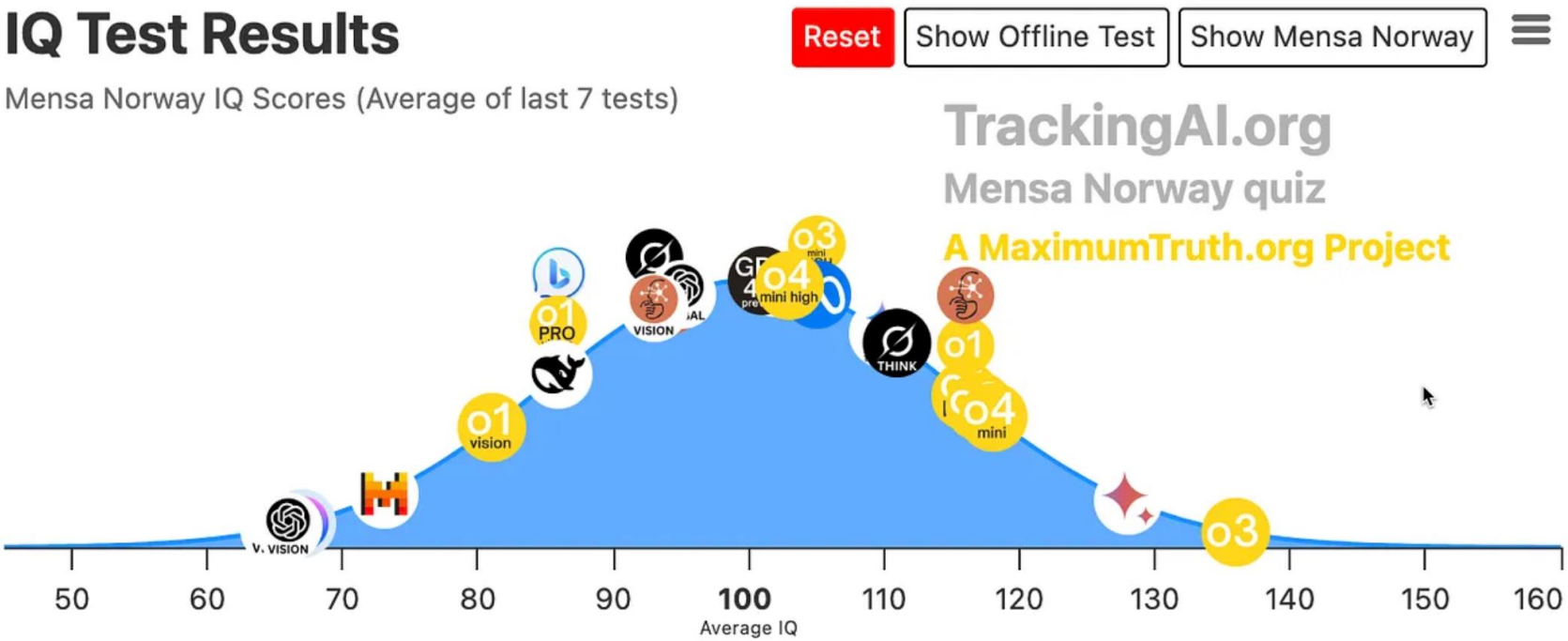
Choosing the right AI tool

- **Coding:** Claude, Grok, ChatGPT, Copilot
- **Research:** ChatGPT, Gemini, Claude, Grok, Deepseek
- **Reasoning:** DeepSeek, Claude, ChatGPT
- **Creative writing:** Claude, ChatGPT, Gemini
- **Current events:** ChatGPT, Grok, Gemini
- **Quick tasks:** Meta AI, ChatGPT


KEEPING UP WITH AI DEVELOPMENTS
AI

AI developments

- **Web search & Reasoning:** In almost all AI tools
- **AI Research Assistants (Deep research):** ChatGPT, Gemini, Grok, Perplexity, Elicit
- **Integrate AI** with Google Workspace, Calendar, Gmail): Claude 4.0
- **AI with Vision:** Share screen, or use a webcam or cell phone camera live (ChatGPT or Gemini)
- **Use in WhatsApp:** Add +1 (833) 436-3285; (Perplexity) & ChatGPT Add +1 800 242 8478 to your contacts
- **Visualise the final product**: Chat GPT's Canvas, Claude's Workspace, Gwen's Lab
- **Google's NotebookLM:** Allows you to upload documents, ask questions, and facilitate conversations between two AI agents.
- **AI Agents:** ChatGPT's Operator (performing tasks online, such as booking trips and purchasing groceries)



From: https://www.maximumtruth.org/p/skyrocketing-ai-intelligence-chatgpts

AI Agents: Open AI Operator or Manus

Operator can handle repetitive browser tasks like filling out forms, comparing options across websites, and performing research (Operator is trained to ask the user to take over when inputting sensitive information like login credentials).

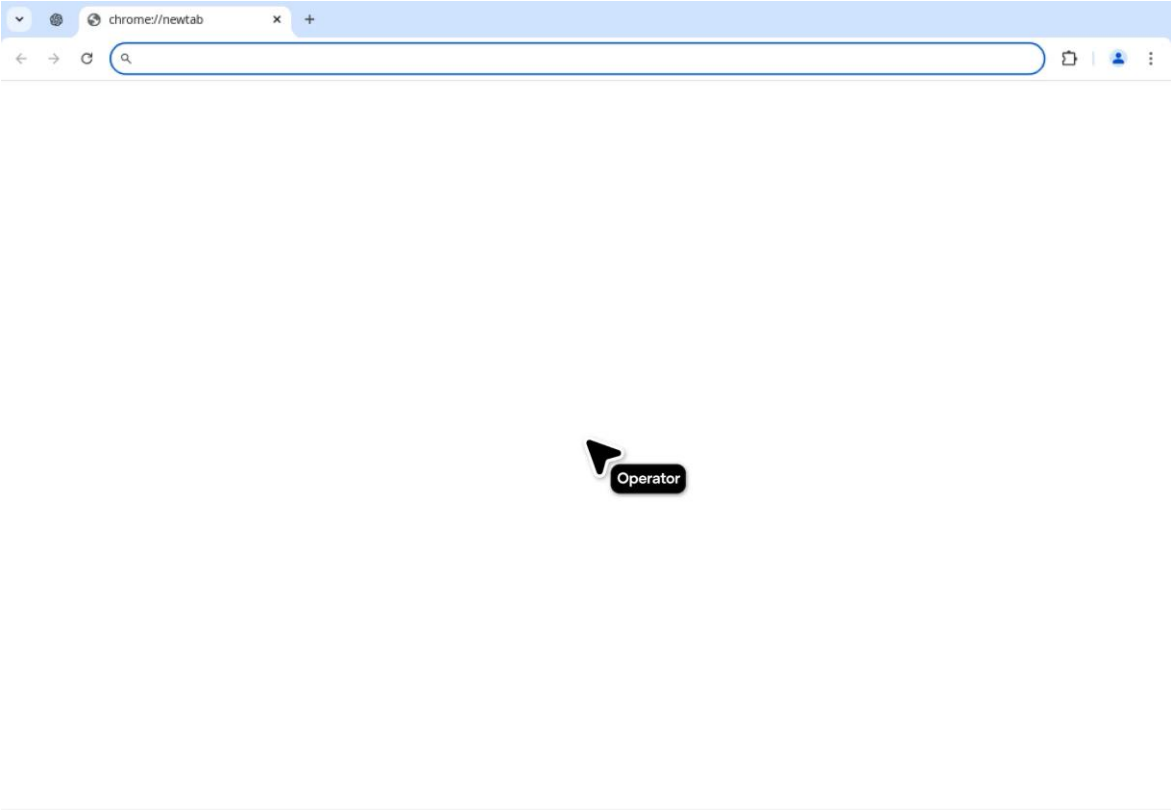
- **Travel Planning:** Plan a [number]-day itinerary from Midrand, South Africa, to [destination] from [start date] to [end date], focusing on [interests, e.g., cultural experiences, adventure activities, relaxation]. Include flight options, accommodation within a [budget] budget, local transportation methods, must-see attractions, dining recommendations (considering [dietary preferences]), and any cultural events happening during my stay.
- **Online shopping:** Search for [specific product] across multiple retailers. Compare prices, shipping times, and key features. Filter for items with at least 4-star reviews and prioritise those with free shipping. Create a comparison of the top three options with their pros and cons, and help me place an order for the best value option (video)
- **Completing online courses:** Operator has been successfully tested completing an Intermediate Python course on Coursera, where it progressed quickly through the course content and even set the video playback to 2x speed. (https://www.classcentral.com/collection/top-free-online-courses)

Example prompt to complete an online course

Complete the online course "Mountains 101" available on Class Central. Follow these steps:

1. Navigate to Class Central and search for "Mountains 101”: https://www.coursera.org/learn/mountains-101
2. Access the course and begin with Lesson 1
3. Watch each video lecture at 1.5x speed
4. Progress through all 12 lessons sequentially
5. Take screenshots of any key diagrams or important slides 6. For each lesson, create a brief summary (2-3 sentences) of the main concepts covered
6. After completing all lessons, compile a list of the most interesting facts learned from the course

If you encounter any quizzes or assessments that are freely accessible, please complete them. Skip any sections that require login credentials or payment. If you encounter any technical difficulties, please let me know.

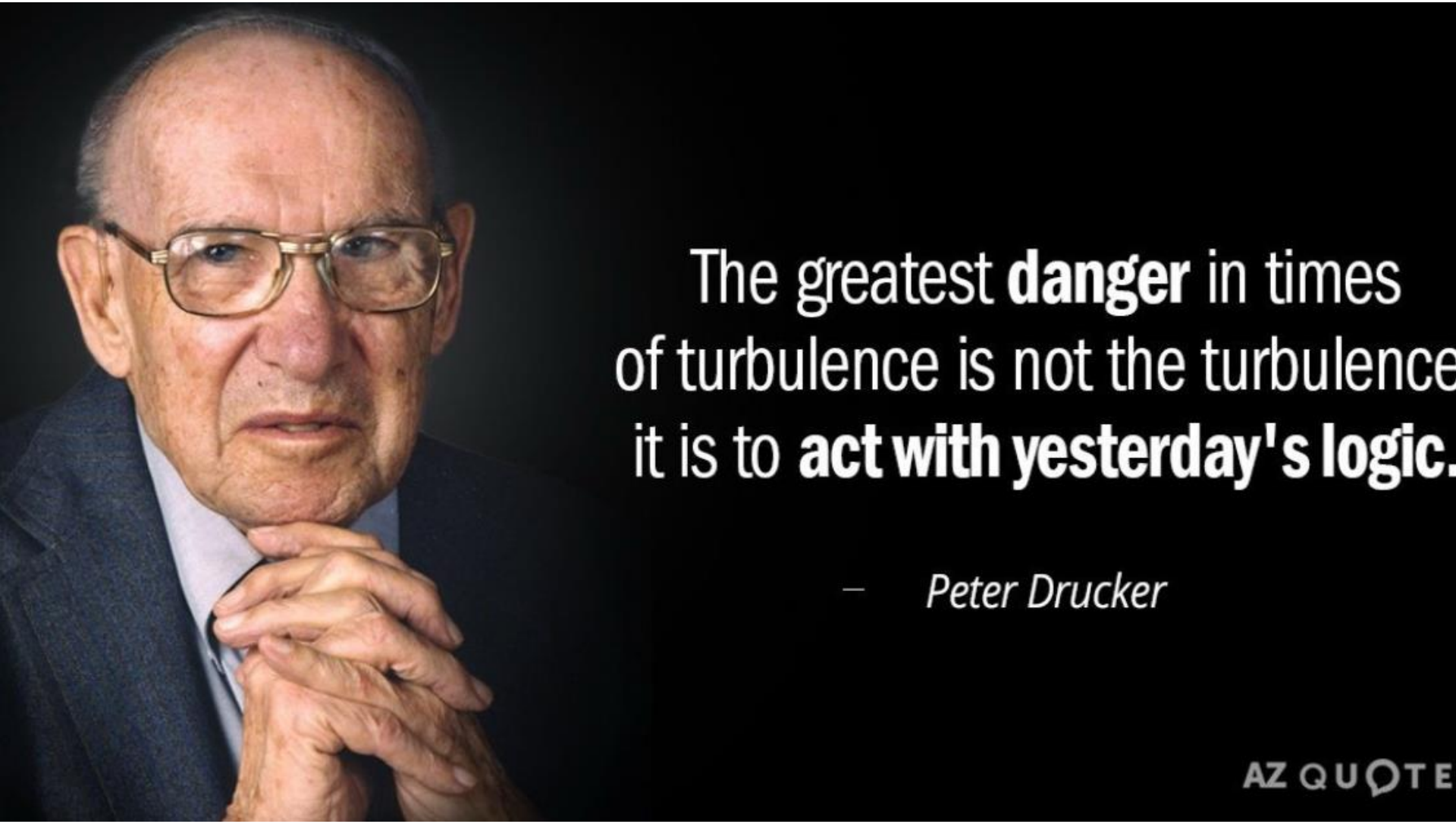


00:00 Navigating to "Mountains 101" course URL

operator.chatgpt.com

Boosting your own efficiency with AI

- Email management
- Research assistant
- Content creation
- Problem-solving partner
- Meeting prep in minutes
- Flowcharts & diagrams
- Image creation
- Data visualisations


The greatest danger in times
of turbulence is not the turbulence
it is to act with yesterday's logic.
— Peter Drucker
AZ QUOTE

How can we integrate AI into Research?

Concerns

Intelligencer SUBSCRIBE

HIGHER EDUCATION

Everyone Is Cheating Their Way Through College ChatGPT has unraveled the entire academic project.

By James D. Walsh, Intelligencer staff writer

14:29 4G

LIVE Following For You

There's a Good Chance Your Kid Uses AI to Cheat

More students are hiding their secret weapon from parents and teachers, racking up top grades while AI chatbots do the work

POV: NSFAS finally releasing funds, you're meeting new people every week , students are no longer hungry , Ai made your degree easier & your group is finished with that annoying project

1,021

6

18

University of Pretoria · City of Tshwane
3830 people shared related posts

Kgotso

I'm not on nsfas #missingmiddle but I
hella heartbreaking to see stude... more

Home Friends Inbox Profile

ChatGPT for Education
37,997 followers
4h • Edited •

From our new report: Students are the # 1 demographic using ChatGPT in the US. Learning and school-related work are their top use cases. ...more

569 35

Universities

UK universities warned to 'stress-test' assessments as 92% of students use AI

Miriam Reyn
Learning design
2d •

As student AI use bec
do about standards cr

Last week Phillip Dawson shared a highly influer
take on the impact of AI on unsupervised asses
https://lnkd.in/gnR8zFgP He suggested that
standards creep was the best case scenario.

I'm not convinced that performance is actually rising as a result of AI use.

But I do think standards in university assessment are already a problem, and that AI will make that problem worse.

#assessment #highereducation #academicintegrity #ai
https://lnkd.in/gEnsc4b6

Soaring standards in the age of AI assessment
themindfile.substack.com

News > Scottish News

PAIN IN THE TECH Scots university students caught using AI to cheat over 1,000 times – and it's 'only the tip of the iceberg'

Students have been using chatbots like ChatGPT and similar software applications to complete coursework and assessments

Paul Rodger, Reporter
Published: 17:05, 2 Mar 2025 | Updated: 17:05, 2 Mar 2025

ChatGPT Search is now free for everyone, no OpenAI account required...
techradar.com

Humanizer.org

Generative AI models like ChatGPT have demonstrated the ability to pass various university-level exams and assessments

- Stanford University - MBA Program (Strategic Management)
- Harvard Law School - JD Program (Constitutional Law)
- Johns Hopkins University - School of Medicine (Medical Physiology)
- Massachusetts Institute of Technology (MIT) - Computer Science and Engineering (Introduction to Algorithms)
- University of Oxford – Humanities (Modern History)

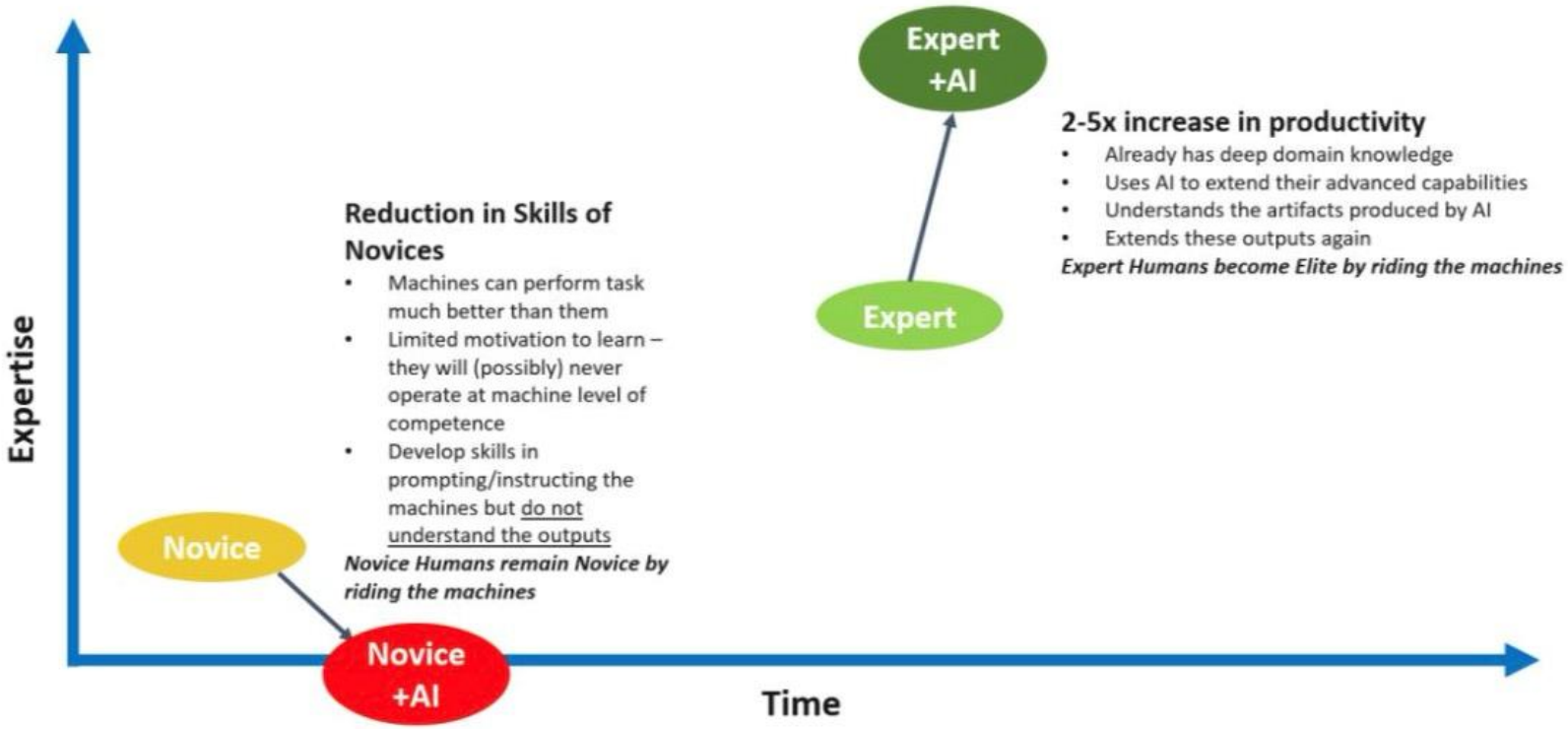
Concerns

The predominant narrative around generative AI in education has focused on concerns related to:

- **University:** Cheating and academic integrity (Damage to the degree's reputation & de-accreditation)
- **Students:** Bypass productive learning, understanding, thinking, knowledge construction (Lacking necessary skills and knowledge may harm future employability opportunities when gaps in competence are exposed in the workplace)

- Generative AI accelerates information access but may undermine foundational knowledge acquisition if misused. Deep learning and internalised foundation knowledge are crucial for critical thinking and problem-solving.
- Deep, well-structured, internalised conceptual knowledge and critical thinking are essential for the effective use of Gen AI.

Figure 7: Experts vs Novices in the World of AI



Ethical guidelines for the use of GenAI and AI • 7 pages

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Original Article

Ethical guidelines for the use of generative artificial intelligence and artificial intelligence-assisted tools in scholarly publishing: a thematic analysis

Adéle da Veiga

School of Computing, College of Science, Engineering and Technology, University of South Africa, Johannesburg, South Africa

Abstract

Purpose: This analysis aims to propose guidelines for artificial intelligence (AI) research ethics in scientific publications, intending to inform publishers and academic institutional policies in order to guide them toward a coherent and consistent approach to AI research ethics.

Methods: A literature-based thematic analysis was conducted. The study reviewed the publication policies of the top 10 journal publishers addressing the use of AI in scholarly publications as of October 2024. Thematic analysis using ATLAS.ti identified themes and subthemes across the documents, which were consolidated into proposed research ethics guidelines for using generative AI and AI-assisted tools in scholarly publications.

Results: The analysis revealed inconsistencies among publishers' policies on AI use in research and publications. AI-assisted tools for grammar and formatting are generally accepted, but positions vary regarding generative AI tools used in pre-writing and research methods. Key themes identified include author accountability, human oversight, recognized and unrecognized uses of AI tools, and the necessity for transparency in disclosing AI usage. All publishers agree that AI tools cannot be listed as authors. Concerns involve biases, quality and reliability issues, compliance with intellectual property rights, and limitations of AI detection tools.

Conclusion: The article highlights the significant knowledge gap and inconsistencies in guidelines for AI use in scientific research. There is an urgent need for unified ethical standards, and guidelines are proposed for distinguishing between the accepted use of AI-assisted tools and the cautious use of generative AI tools.

Keywords

Artificial intelligence; Authorship; Organizational policy; Scholarly communication; Generative artificial intelligence

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Kirsten Krauss, LinkdIN

Polluted Science in GenAI tools for literature reviews

Assessing the 'Top 6 Papers' returned by five different tools

Study field:	SCISPACE	ELICIT	WEB OF SCIENCE RESEARCH ASSISTANT	LITMAPS	SEMANTIC SCHOLAR
Pharmacology	- **2 usable journal articles** - **3 predatory journals** - **1 unknown/ unlisted journal** 2 / 4	- **6 usable journal articles** 6 / 0	- **5 usable journal articles** - **1 predatory publisher** 5 / 1	- **5 usable journal articles** - 1 unknown publication 5 / 1	Only 2 results: - **1 usable journal article** - **1 unknown/ unlisted title** 1 / 1
Psychology	- **4 usable journal articles** - 1 dissertation - 1 dataset 6 / 0	- **6 usable journal articles** 6 / 0	- **6 usable journal articles** 6 / 0	- **6 usable journal articles** 6 / 0	- **6 usable journal articles** 6 / 0
Information Systems	- **5 usable references, only 1 journal article** - **1 unknown/ unlisted journal** 5 / 1	- **1 usable journal article** - 5 conference papers 6 / 0	- **5 usable journal articles** - 1 conference paper 6 / 0	- **4 usable journal articles** - 2 conference publications 6 / 0	- **3 usable journal titles** - **1 predatory journal** - 1 conference paper - 1 book chapter 5 / 1
Bioengineering	- **3 usable journal articles** - **1 unknown/ unlisted journal** - **2 predatory publishers** 3 / 3	- **1 journal article** - **1 predatory journal** - **1 predatory publisher** - **3 unknown/ unlisted journals** 1 / 5	- **3 usable journal articles** - **3 predatory publishers** 3 / 3	Only 1 result: - **1 usable journal article** 1 / 0	- **6 usable journal articles** 6 / 0

Generative AI in Master Research:

Literature Review Assistance

- Summarise and compare key IS frameworks, methodologies, and academic studies.
- Generate annotated bibliographies on topics such as *cloud adoption strategies, enterprise architecture models, or data governance*.

Hypothesis Formation & Conceptual Exploration

- Explore new IS theories or conceptual models.
- Suggest alternative interpretations of existing frameworks (e.g., differing views on *business–IT alignment* or *digital transformation maturity*).

Drafting Support

- Assist in drafting research proposals, problem statements, and objectives.
- Produce first-pass outlines for IS artefacts, evaluation frameworks, or policy recommendations for organisations adopting new technologies.

Generative AI in Master Research

Data Analysis

- Analyse large volumes of system logs, enterprise data, or survey results.
- Identify patterns in IT adoption, user behaviour, or system performance.

Comparative & Interdisciplinary Research

- Compare Information Systems frameworks, architectures, or standards using summarisation.
- Cross-reference with insights from business management, economics, psychology, or ethics to broaden IS research.

Some tools:

- Literature Research:
  - Non_AI: Google Scholar
  - Elicit
  - Scite.ai Not just references, but “smart citations”: tells you whether later work supports, disputes, or merely mentions a paper.
  - Iris.ai – Maps out research questions into clusters of literature.
  - Perplexity.ai (academic mode) – Conversational search that can cite sources live, sometimes bridging grey literature and published work.

When to use what:

Discovery & Scoping – *Perplexity.ai*

Use conversational queries to quickly surface recent cases, policy briefs, NGO reports, and relevant statutes.

Great for scanning the horizon and pulling in grey literature.

Deep Academic Context – *Semantic Scholar*

Search for peer-reviewed articles and legal scholarship.

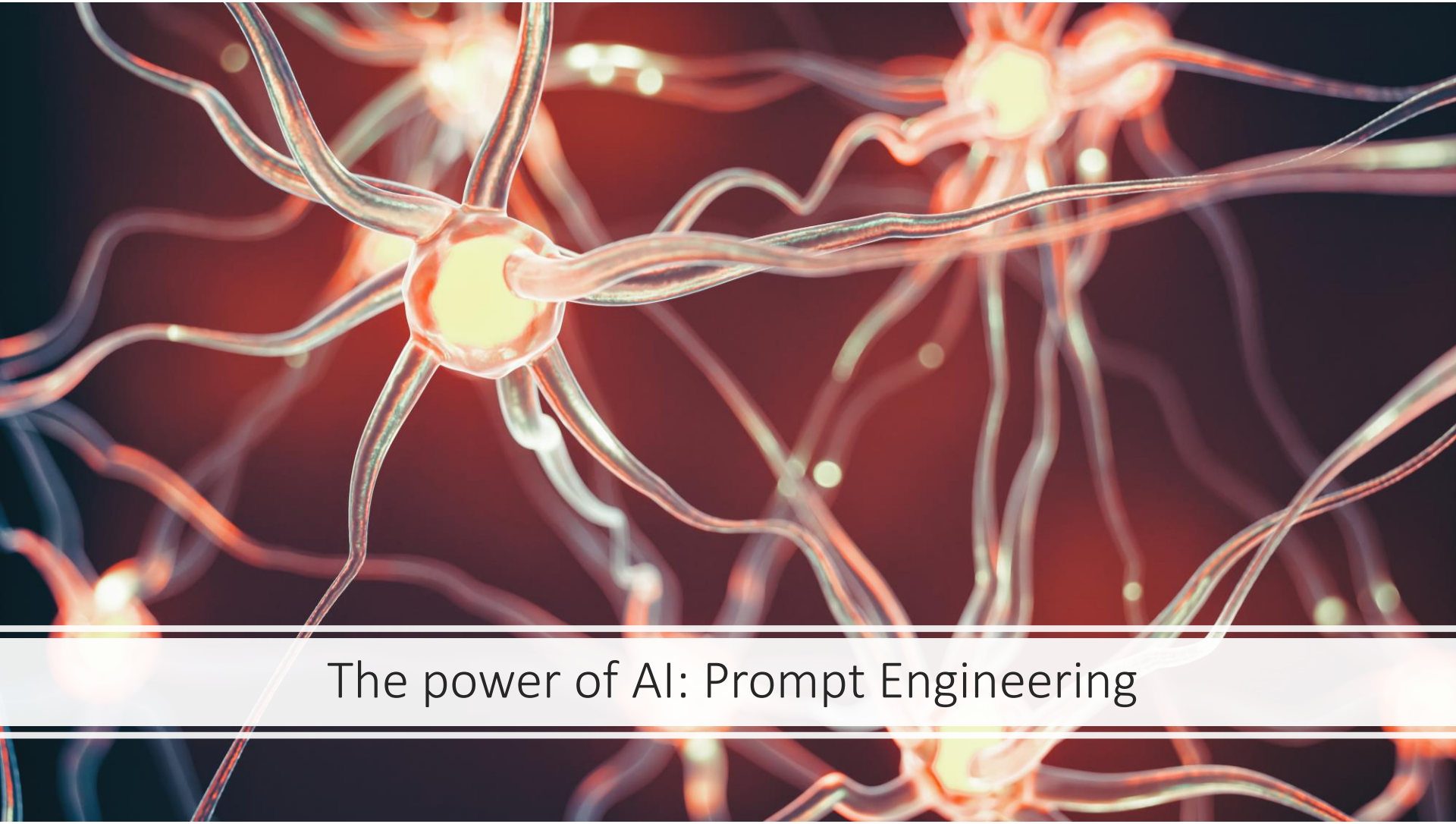
Use citation graphs to trace influential authors, theories, and precedent analysis.

Evidence Strength Check – *Scite.ai*

Take the key articles you found and check how later scholarship treats them.

See if arguments are supported, disputed, or just cited in passing.

Think of it like a funnel: Perplexity for breadth → Semantic Scholar for depth → Scite for reliability.



The power of AI: Prompt Engineering

Prompt Engineering for GenAI

1. Frame prompts as “what if” explorations, not answer requests.
Instead of: *“Summarise this case.”*
Try: *“What if I read this case through a critical legal studies lens? How would the interpretation differ from a positivist reading?”*
→ This nudges the AI into contrasting perspectives, helping you discover blind spots.

Prompt Engineering for GenAI

2. Force multiple framings in a single prompt.
Ask the AI to generate **three alternative explanations** or **contrasting hypotheses**.
E.g., *"Give me three different theoretical framings of data privacy regulation: one economic, one ethical, one jurisprudential."*
→ This builds intellectual breadth, not tunnel vision.

Prompt Engineering for GenAI

3. Be explicit about the role you want the AI to play.
Treat it like a seminar partner, devil's advocate, or novice student.
E.g., *"Critique my argument as if you were an examiner looking for weaknesses in my methodology."*
→ This makes the AI less of a secretary and more of a sparring partner.

Prompt Engineering for GenAI

4. Layer prompts iteratively.
Don't expect brilliance in one shot. Use **chains of refinement**:

- Step 1: "Sketch 5 possible angles to study judicial bias in sentencing."
- Step 2: "Take angle #2 and expand it into a possible research question with pros/cons."
- Step 3: "Suggest methods that could test this."
  → Iteration = depth.

Prompt Engineering for GenAI

5. Demand evidence and counter-evidence.
Instead of just "summarise," push it to **argue both sides**.
E.g., *"Present arguments for and against the claim that AI-driven contract analysis reduces systemic bias. Add references or examples where possible."*
→ This cultivates the academic habit of critical balance.

Cheat Card: Prompt Engineering for Master's Research

Using Generative AI as a Thinking Partner

1. Frame prompts as 'what if' explorations.

→ Ask the AI to reinterpret your research problem using different practical or theoretical approaches.

2. Use multiple framings in one prompt.

→ Request 2–3 alternative explanations, models, or solutions relevant to your topic.

3. Be clear about the role the AI should play.

→ Use roles like supervisor, peer reviewer, or practitioner in the field.

4. Layer prompts step by step.

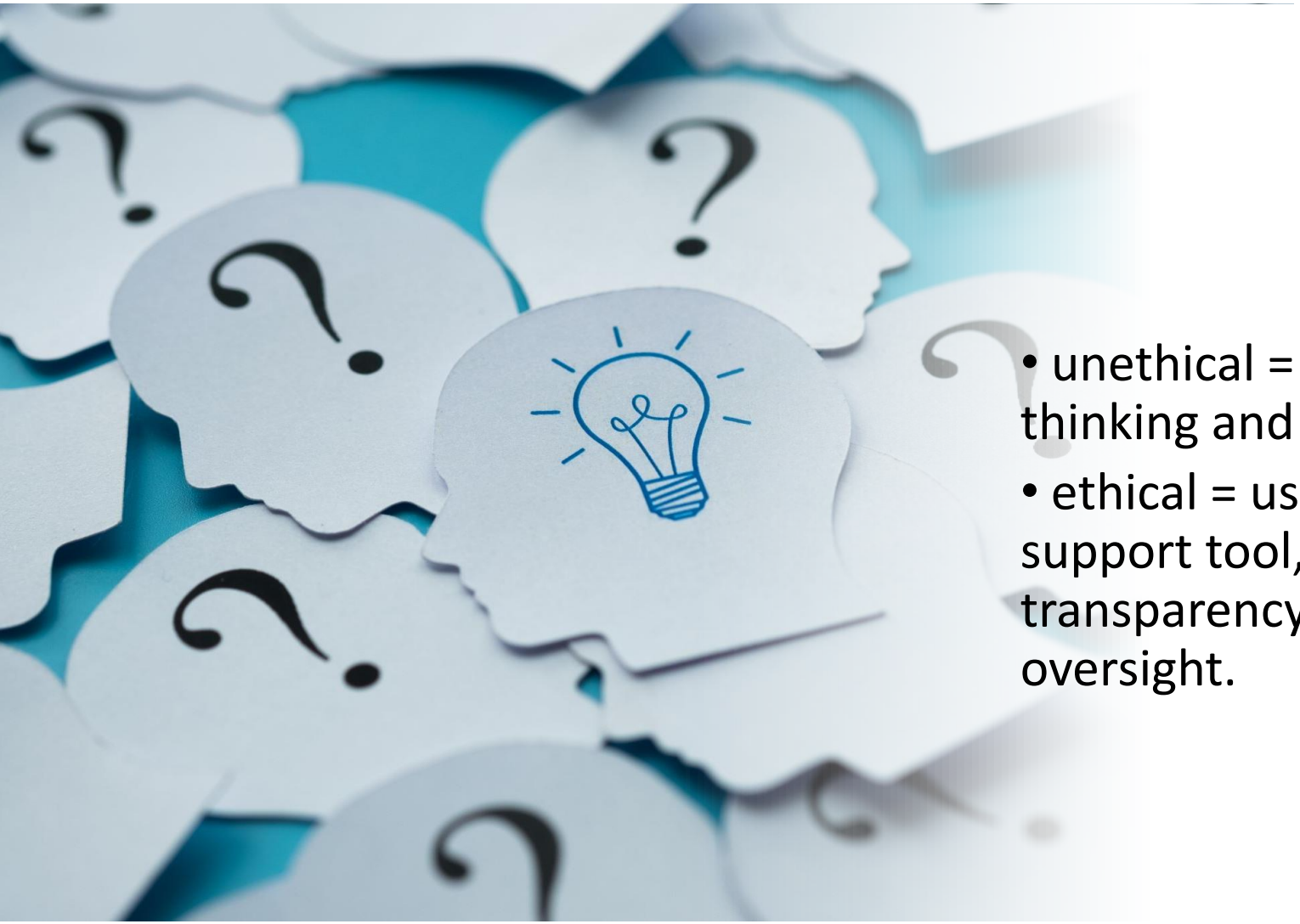
→ Start broad with your research area, then refine toward your specific research questions or methods.

5. Request support for arguments and counterarguments.

→ Always ask for both strengths and weaknesses of an approach, with examples tied to Information Systems.

An AI research Bot: Creator Alta vd Merwe

- https://chatgpt.com/g/g-684a9be536e481918fe581c62d7fc397-ebit-research-mentor



- unethical = outsourcing thinking and hiding it;
- ethical = using AI as a support tool, with transparency and critical oversight.

Unethical Use of AI Tools in PhD Research	Ethical Use of AI Tools in PhD Research
Submitting AI-generated text as original work without disclosure (plagiarism).	Using AI to brainstorm, outline, or structure ideas, while making the final work your own and acknowledging support.
Fabricating references, data, or citations generated by AI.	Using AI to help discover real references more efficiently, then verifying them through academic databases.
Letting AI write full thesis chapters without critical engagement.	Using AI as a writing assistant to improve clarity, grammar, or style, while ensuring the intellectual contribution is yours.
Using AI to bypass originality checks or disguise copied content.	Employing AI to check for unintentional plagiarism and improve paraphrasing ethically.
Relying on AI to generate false survey or experimental results.	Applying AI to analyse real data sets, visualise findings, or run simulations with transparency.
Misrepresenting AI-produced insights as your own novel theories.	Using AI to compare, synthesise, and critically engage with existing literature.
Not disclosing AI involvement in research or writing.	Clearly stating in methodology or acknowledgements how AI tools were used.
Using AI to impersonate peer reviewers or manipulate the review process.	Using AI to practise peer-review skills (e.g., asking AI to critique your draft as a "mock reviewer").

Where do I find AI tools?

https://theresanaiforthat.com/

Go to our Spreadsheet;

Do the following exercise and copy your answers into your TAB;

Generative AI

- Open ChatGTP (https://chat.openai.com/) and give it your Main research question.
- Ask ChatGTP how can you refine this research question to be more exact (PUT YOUR RESULTS IN YOUR TAB).
- Key Findings from AI: (PUT YOUR RESULTS IN YOUR TAB).
- Query for possible Sub-Questions: (PUT YOUR RESULTS IN YOUR TAB).
- Which of these sub-questions make sense to use? (PUT YOUR RESULTS IN YOUR TAB).

On using OpenAI such as ChatGPT

- One way to reference it:
- Enterprise Architecture:
  - Enterprise Architecture (EA) is a discipline that involves creating and managing a comprehensive and integrated view of an organization's people, processes, information, and technology, and the relationships between them, to guide the organization's strategy, goals, and operations towards achieving its objectives. It provides a structured approach for designing and evolving the organization's architecture to support its current and future business needs, while optimizing its resources, minimizing risk, and improving its overall performance (ChatGPT, 2023).
  - ChatGPT (2023) Prompt: Give a definition for Enterprise Architecture, accessed 20 March 2023;

Record your ideas ..Your ideas will become key concepts in your dissertation!

To session 1.7